Shaowen Wang

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Research Interests

Differentiable wave-physics solvers and implicit neural representations (INRs) for full-waveform inversion (FWI) and seismic imaging; deep learning for seismic data processing; GPU and high-performance computing.

Experience

KAUST, Postdoctoral Fellow, Physical Science and Engineering Division — Thuwal, Saudi Arabia — 2024 – present

- Differentiable wave-physics solvers and implicit neural representations for FWI and seismic imaging
- Author of SWEEP (differentiable wave simulation framework) and IFWI (implicit FWI); GPU and HPC for large-scale inversion

Education

PhD	**Ocean University of China**, Marine Geophysics	2020 – 2024
MSc	**Ocean University of China**, Geo-detection and Information Technology	2018 – 2020
BSc	**Shandong University of Science and Technology**, Prospecting Technology and Engineering	2014 – 2018

Journal Articles

L. Wang, **Shaowen Wang**, F. Min, T. Alkhalifah (2026). **G²IFWI: Gradient-Coupled Diffusion Priors for Regularizing Implicit Full Waveform Inversion**. *Journal of Geophysical Research: Machine Learning and Computation*.

Shaowen Wang, T. Alkhalifah (2026). **Implicit Full Waveform Inversion Imaging**. *Geophysical Journal International*.

Shaowen Wang, J. Tan, P. Song, B. He, Q. Wang, G. Du (2024). **Deep Learning for Noise Attenuation from the Ocean Bottom Node 4C Data**. *Geophysical Prospecting*.

J. Tan, J. Wang, P. Song, **Shaowen Wang**, D. Xia, G. Du, Q. Wang (2024). **Surface-related Multiple Prediction for Ocean-bottom Node Data Based on Demigration using Downgoing Wave Imaging Data**. *Geophysical Prospecting*.

Shaowen Wang, Y. Jiang, P. Song, J. Tan, Z. Liu, B. He (2023). **Memory Optimization in RNN-based Full Waveform Inversion using Boundary Saving Wavefield Reconstruction**. *IEEE Transactions on Geoscience and Remote Sensing*.

Shaowen Wang, P. Song, J. Tan, B. He, D. Xia, Q. Wang, G. Du (2023). **Deep Learning-based Attenuation for Shear-wave Leakage from Ocean-bottom Node Data**. *Geophysics*.

Shaowen Wang, P. Song, J. Tan, D. Xia, G. Du, Q. Wang (2023). **Deblending of Seismic Data in the Wavelet Domain via a Convolutional Neural Network Based on Data Augmentation**. *Geophysical Prospecting*.

Q. Q. Wang, P. Song, Q. F. Hua, B. H. Liu, G. B. Li, **Shaowen Wang**, G. N. Du (2023). **Full Waveform Inversion Based on Adam Algorithm with Optimized Parameters**. *Chinese Journal of Geophysics*.

Shaowen Wang, P. Song, J. Tan, B. He, Q. Wang, G. Du (2022). **Attention-based Neural Network for Erratic Noise Attenuation from Seismic Data with a Shuffled Noise Training Data Generation Strategy**. *IEEE Transactions on Geoscience and Remote Sensing*.

Shaowen Wang, P. Song, J. Tan, C. Xie, B. Zhao, S. B. Mao (2022). **The Least-squares Reverse Time Migration with Gradient Optimization Based on QHAdam**. *Chinese Journal of Geophysics*.

C. Xie, P. Song, Z. H. Zou, J. Tan, **Shaowen Wang**, B. Zhao (2022). **Cross-correlation Weighted Reverse Time Migration with Wavefield Decomposition Based on Optical Flow Vector**. *Chinese Journal of Geophysics*.

D. Xia, P. Song, X. Li, J. Tan, C. Xie, **Shaowen Wang**, K. Liu, B. Zhao, S. Mao (2022). **Time Domain Full Waveform Inversion Based on Gradient Preconditioning with an Angle-dependent Weighting Factor**. *Journal of Ocean University of China*.

Shaowen Wang, P. Song, J. Tan, C. Xie, S. Mao, Q. Wang (2021). **Gaussian-type Weighted Hybrid Absorbing Boundary for Elastic Wave Simulation and its Acceleration on GPU**. *Oil Geophysical Prospecting*.

C. Xie, P. Song, X. Li, J. Tan, **Shaowen Wang**, B. Zhao (2021). **Angle-weighted Reverse Time Migration with Wavefield Decomposition Based on the Optical Flow Vector**. *Frontiers in Earth Science*.

C. Xie, P. Song, X. Li, J. Tan, **Shaowen Wang**, B. Zhao (2021). **Least-squares Reverse Time Migration using the Gradient Preconditioning Based on Transmitted Wave Energy**. *Frontiers in Earth Science*.

R. Zhang, P. Song, B. Liu, X. Zhang, J. Tan, Z. Zou, C. Xie, **Shaowen Wang** (2020). **Low-frequency Swell Noise Suppression Based on U-Net**. *Applied Geophysics*.

Conference Proceedings

Shaowen Wang, T. Alkhalifah (2026). **Self-supervised DAS Data Lossy Compression Based on Anisotropic Multi-resolution Hash Encoding**. *International Meeting for Applied Geoscience & Energy (IMAGE)*.

L. Wang, **Shaowen Wang**, F. Min, T. Alkhalifah (2026). **Learned Regularizations for Hash-encoding Implicit FWI using Well-constrained Diffusion Models**. *International Meeting for Applied Geoscience & Energy (IMAGE)*.

L. Wang, O. M. Saad, **Shaowen Wang**, F. Min, T. Alkhalifah (2026). **Seismic Data Interpolation using Neural Representations with Hybrid Fourier–Hash Encoding**. *International Meeting for Applied Geoscience & Energy (IMAGE)*.

Shaowen Wang, T. Alkhalifah (2026). **Pretraining for Faster Implicit Full Waveform Inversion under Multiresolution Hash Encoding**. *87th EAGE Annual Conference & Exhibition*.

Shaowen Wang, T. Alkhalifah (2026). **Spectral Warm-up for Neural-Reparameterized Full Waveform Inversion**. *87th EAGE Annual Conference & Exhibition*.

L. Wang, O. M. Saad, **Shaowen Wang**, F. Min, T. Alkhalifah (2026). **A Hybrid Fourier-Hash Encoded Implicit Neural Representations for Seismic Data**. *87th EAGE Annual Conference & Exhibition*.

L. Wang, **Shaowen Wang**, F. Min, T. Alkhalifah (2026). **Learned Regularizations for Multiresolution Hash Encoding-based Implicit Full Waveform Inversion Using Diffusion Models**. *87th EAGE Annual Conference & Exhibition*.

Shaowen Wang, T. Alkhalifah (2026). **Waveform Inversion and Imaging using Implicit Neural Representations**. *Sixth EAGE Digitalization Conference & Exhibition*.

Shaowen Wang, T. Alkhalifah (2025). **Accelerating the Convergence of Implicit FWI and LSRTM with a Field Data Application**. *International Meeting for Applied Geoscience & Energy (IMAGE)*.

Shaowen Wang, M. Ravasi, T. Alkhalifah (2025). **Multiresolution Hash Encoding for High Resolution Implicit Full Waveform Inversion**. *86th EAGE Annual Conference & Exhibition*.

Shaowen Wang, T. Alkhalifah (2025). **Implicit Full Waveform Inversion with Energy-Weighted Gradient**. *86th EAGE Annual Conference & Exhibition*.

Shaowen Wang, P. Song, J. Tan, D. Xia, B. Zhao, S. Mao (2024). **Differentiable Traveltime Misfit for Wave-Equation Tomography**. *85th EAGE Annual Conference & Exhibition*.

Preprints

L. Wang, **Shaowen Wang**, F. Min, T. Alkhalifah (2026). **Coarse-to-fine Multi-resolution Hash Encoding for Implicit Full Waveform Inversion**. *arXiv:2610.01081*.

Shaowen Wang, T. Alkhalifah (2026). **SWEEP (Seismic Wave Equation Exploration Platform): A Unified Solver Framework for Differentiable Wave Physics**. *arXiv:2604.14189*.

Shaowen Wang, T. Alkhalifah (2026). **Target Noise: A Pre-training Based Neural Network Initialization for Efficient High Resolution Learning**. *arXiv:2602.06585*.

Skills

Programming: Python, PyTorch, CUDA, C/C++

Geophysics: FWI, RTM/LSRTM, wave-equation modeling, OBN and marine seismic data processing

Computing: GPU acceleration, multi-GPU and distributed computing (HPC)